Cambridge (CIE) A Level Maths: Pure 1

Your notes

Circular Measure (Radians)

Contents

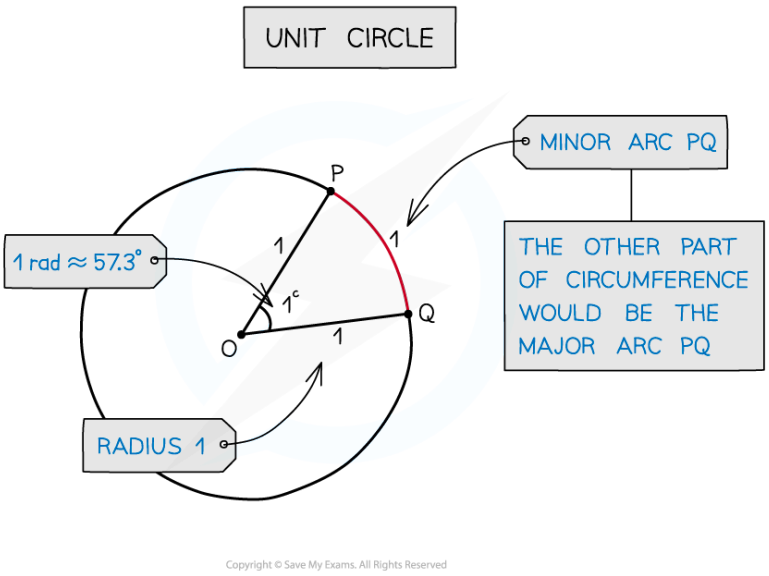
- Radian Measure
- Trigonometry Exact Values

Your notes

Radian measure

What are radians?

- **Radians** are an alternative to degrees for measuring angles

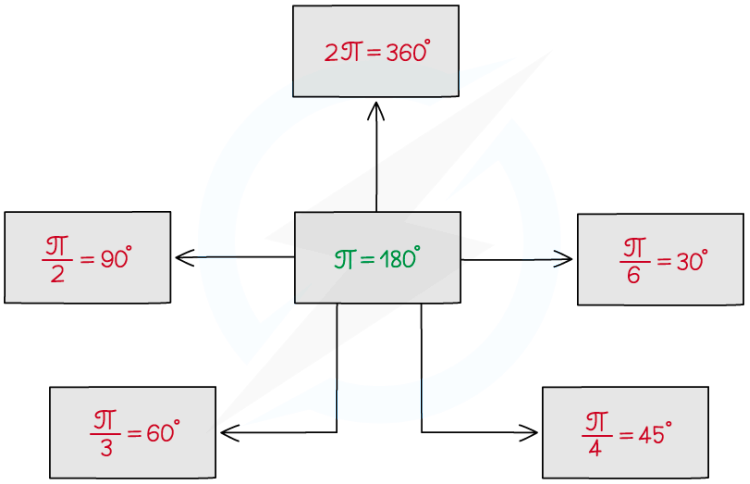


- 1 radian is the angle in a sector of radius 1 and arc length 1
- Radians are normally quoted in terms of π
- This leads to
  - **2π = 360°**
  - **π = 180°**
- The symbol for radians is ^{c} but it is more usual to see **rad**
  - Often, when π is involved, no symbol is given as it is obviously in radians
- Radians are used in **trigonometry** (see Exact Values) and **calculus** (First Principles Differentiation – Trigonometry)

How do I convert between radians and degrees?

- The most common are below, these are helpful to know

Your notes



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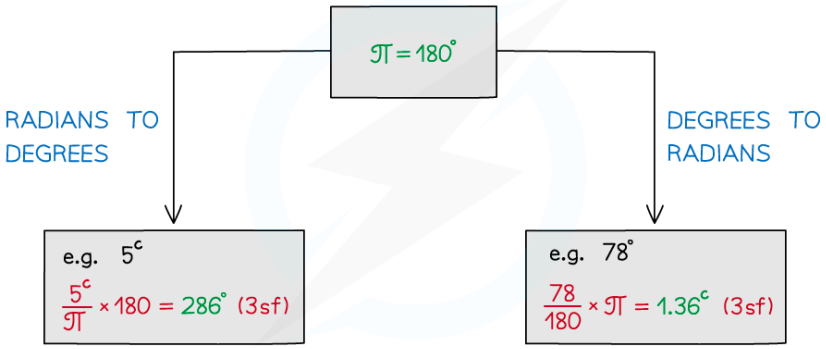
- Multiples of these are quite common too, eg ¾π = 135°
- For less familiar angles use **π = 180°** to convert



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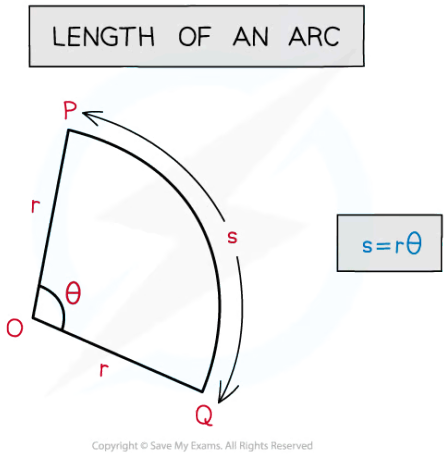
How do I find the arc length of a sector with radians?



- The length of an arc is **s = rθ**

e.g. FIND THE LENGTH OF THE ARC PQ

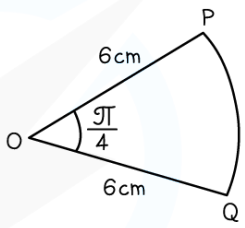
"s" = PQ

"r" = 6

"θ" = $\frac{\pi}{4}$

∴ PQ = $6 \times \frac{\pi}{4}$

PQ = $\frac{3}{2}\pi$ cm



ANSWER IN TERMS OF π (UNLESS ASKED OTHERWISE)

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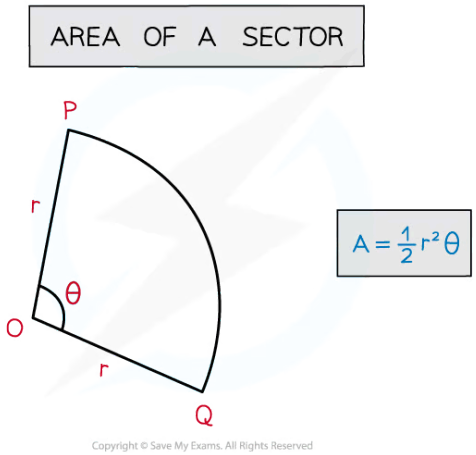
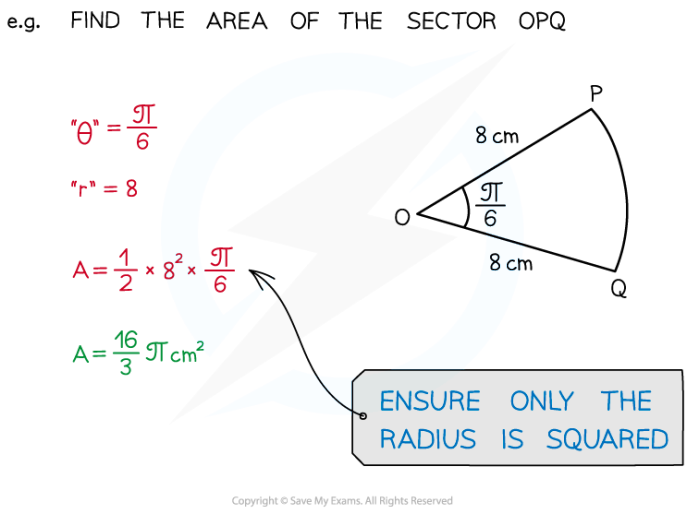
How do I find the area of a sector with radians?

Your notes



- The area of a sector is **A = ½r²θ**

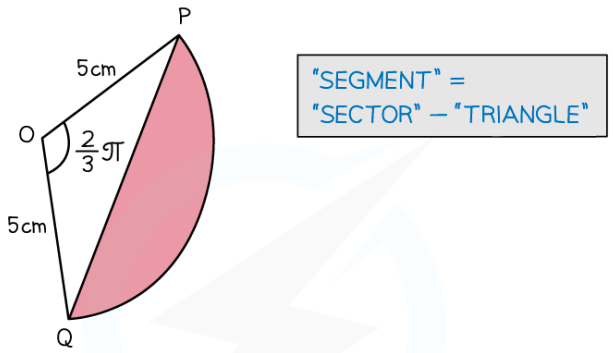


How do I solve sector problems with radians?

- Other **angle**, **circle**, **area**, etc skills may be needed
- For example, area of a triangle **"A = ½absinC"**

Your notes

e.g. FIND THE AREA OF THE SHADED SEGMENT



AREA OF SECTOR OPQ $= \frac{1}{2} \times 5^2 \times \frac{2}{3}\pi = \frac{25}{3}\pi$ "$\frac{1}{2}r^2\theta$"

AREA OF TRIANGLE OPQ $= \frac{1}{2} \times 5 \times 5 \times \sin\frac{2}{3}\pi = \frac{25\sqrt{3}}{4}$ "$\frac{1}{2}ab\sin C$"

$\therefore$ AREA OF SEGMENT $= \frac{25}{3}\pi - \frac{25\sqrt{3}}{4}$

$= \frac{100\pi - 75\sqrt{3}}{12}\ \text{cm}^2$

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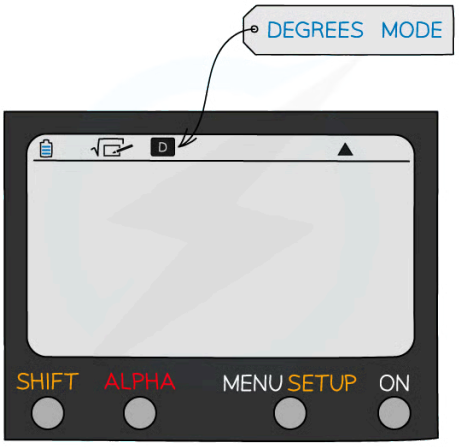
Examiner Tips and Tricks

- Exam papers will have a mixture of questions in degrees and radians
- Check the mode (degrees/radians) of your calculator before using it
- Ensure you know how to change the mode on your calculator
- For the Casio fx-991EX Classwiz
  - **SHIFT** then **MENU SETUP**
  - Choose option **2: ANGLE UNIT**
  - Choose either **1: DEGREE** or **2: RADIAN**

Your notes



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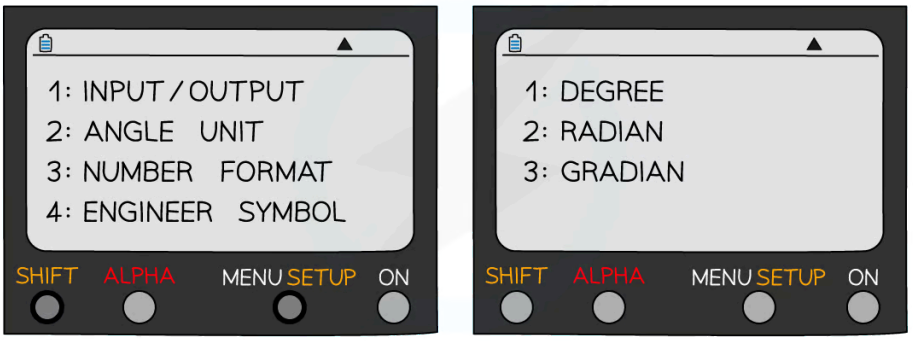

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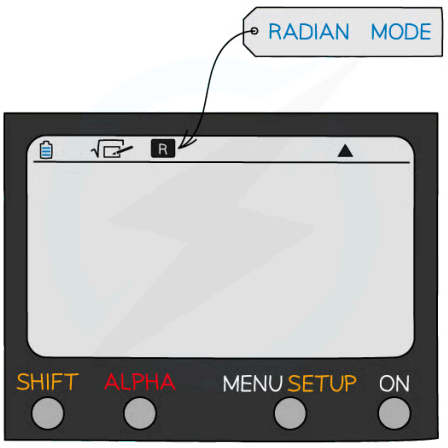

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Worked Example

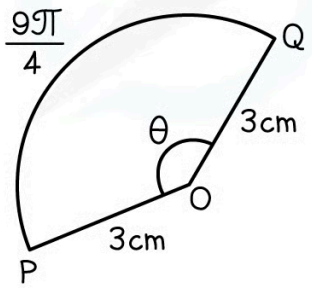
Your notes

In the diagram below OPQ is the sector of a circle with radius 3 cm.

The length of the arc PQ is $\frac{9\pi}{4}$.

The angle at the centre of the circle is θ, measured in radians.

Find the area of the sector OPQ.



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"$A = \frac{1}{2}r^2\theta$"

"r" = 3

$\theta = ?$

USE LENGTH OF ARC TO FIND θ

"$s = r\theta$"

$\frac{9\pi}{4} = 3\theta$

$\theta = \frac{3\pi}{4}$

∴ AREA OF SECTOR OPQ $= \frac{1}{2} \times 3^2 \times \frac{3\pi}{4}$

$= \frac{27\pi}{8}\ \text{cm}^2$

NOW FIND THE AREA USING θ AND r

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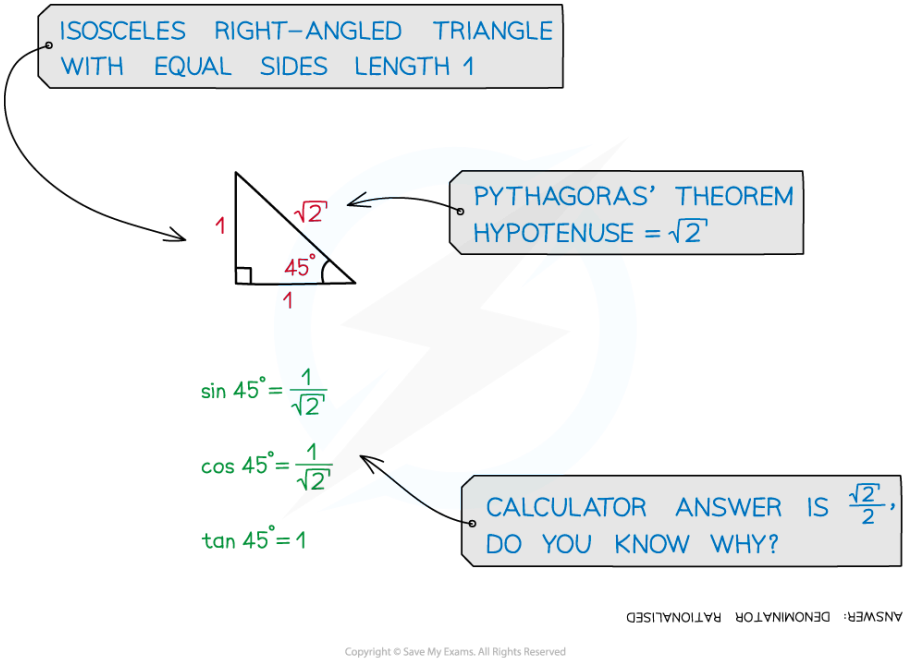
Your notes

Trigonometry exact values

What are exact values in trigonometry?

- In trigonometry some values of sin, cos and tan are "nice"
- For example **sin 30° = ½**, **tan 45° = 1**
- We can find those associated with the angles **30°**, **45°** and **60°** using SOHCAHTOA on isosceles and equilateral triangles



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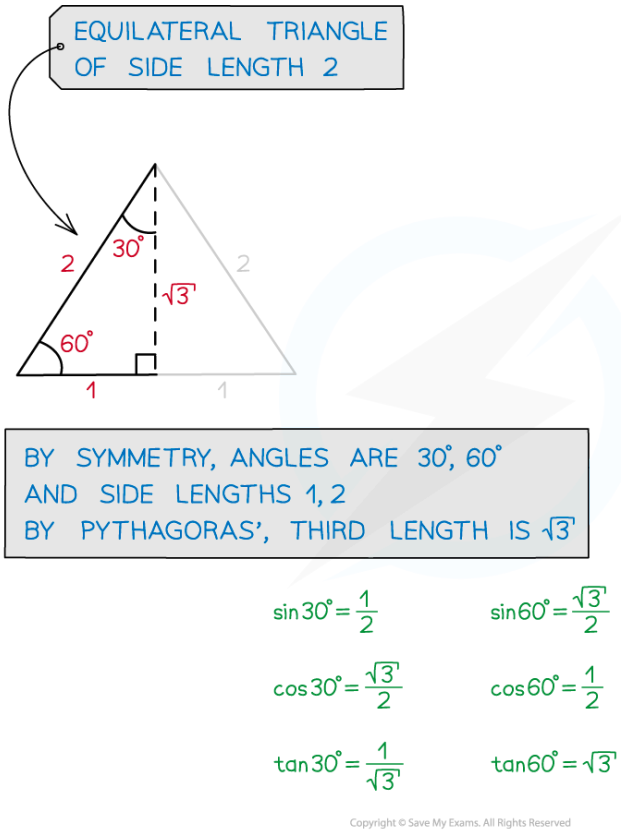

- **sin**, **cos** and **tan** values for **30°**, **45°** and **60°** should be recalled easily
- Memorise either the actual values or how to work them out from the triangle
- **sin**, **cos** and **tan** for **0°**, **90°**, **180°** should also be very familiar
- They can be recalled using the relevant **trigonometric graph** (see Graphs of Trigonometric Functions)

What exact values do I need to know in trigonometry?

- All of the above applies to radians as well as degrees
- Here is a table of all exact values including **radians** (see Radian Measure)

Your notes

Your notes

DEGREES	0°	30°	45°	60°	90°	180°	360°
RADIANS	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	2π
sin	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	0	0
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	−1	1
tan	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	UNDEFINED	0	0

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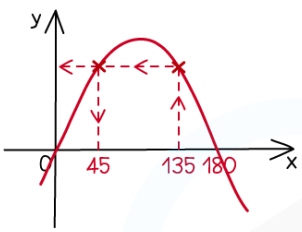
How do I find exact values of other angles?

- Exact **sin**, **cos** and **tan** values of **multiples** of **30°**, **45°**, **60°** can also be found
- This is a combination of recalling the basic values and using the graph

e.g. FIND THE EXACT VALUE OF $\sin 135^\circ$

STEP 1



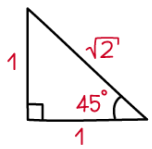
SKETCH THE GRAPH OF sinx

STEP 2

$\sin 135^\circ = \sin 45^\circ$

SYMMETRY OF GRAPH RELATES 135° TO 45°

STEP 3



$\sin 45^\circ = \frac{1}{\sqrt{2}}$

RECALL EXACT VALUE
DRAW DIAGRAM TO HELP

$\therefore \sin 135^\circ = \frac{1}{\sqrt{2}}$

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Your notes

Examiner Tips and Tricks

- Draw the triangles for **sin**, **cos** and **tan** of **30°**, **45°** and **60°** on the exam paper so that you can then use them as many times as you need throughout the paper.

Worked Example

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Your notes

Write down the exact values of the following:

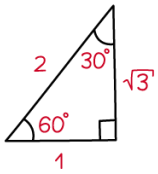
(i) $\sin 30°$

(ii) $\cos\frac{\pi}{3}$

(iii) $\tan 135°$

(iv) $\sin\frac{4\pi}{3}$

(i)



USE TRIANGLE TO RECALL EXACT VALUE

$\sin 30° = \frac{1}{2}$

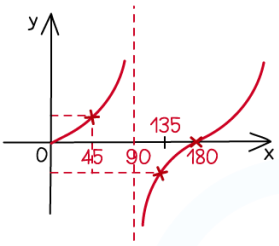
(ii) $\frac{\pi}{3} = 60°$

USE DIAGRAM ABOVE

$\cos\frac{\pi}{3} = \frac{1}{2}$

(iii) STEP 1

SKETCH GRAPH OF tanx

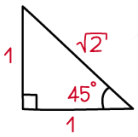


STEP 2

USE SYMMETRY OF GRAPH

$\tan 135° = -\tan 45°$

STEP 3



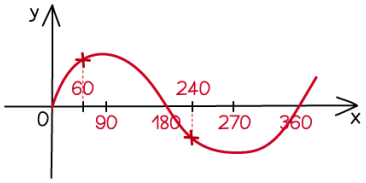
RECALL EXACT VALUE

$\tan 135° = -1$

(iv) $\frac{4\pi}{3} = 4\left(\frac{\pi}{3}\right) = 4 \times 60 = 240°$

CONVERSION NOT ESSENTIAL

STEP 1



SKETCH THE GRAPH OF sinx

STEP 2

$\sin 240^\circ = -\sin 60^\circ$

SYMMETRY

$\sin \frac{4\pi}{3} = -\sin \frac{\pi}{3}$

STEP 3

$\sin \frac{4\pi}{3} = -\frac{\sqrt{3}}{2}$

USING TRIANGLE FROM PART (i)

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